Year 10 Foundation

# REVISION PACK

## Progress Check 1

Thursday 1 October 2026

**Name:** _____

**Class:** _____

### Units in this pack

- Number 1
- Algebra 1

## Number 1 · Working with fractions (Lesson 1.1)

**Worked example: adding fractions**

Work out $\frac{2}{3} + \frac{1}{4}$

| | |
|---|---|
| **Step 1** Same bottom | $\frac{8}{12} + \frac{3}{12}$ |
| **Step 2** Add the tops | $\frac{11}{12}$ |
| **Step 3** Answer | $\frac{11}{12}$ |

### Practice: adding fractions

**1.1** Work out $\frac{3}{5} + \frac{1}{4}$

**1.2** Work out $\frac{5}{6} + \frac{1}{8}$

**1.3** Work out $\frac{7}{10} + \frac{1}{6}$

**1.4** Work out $\frac{2}{7} + \frac{1}{3}$

**1.5** Work out $\frac{1}{2} + \frac{2}{9}$

**1.6** Work out $\frac{3}{8} + \frac{1}{3}$

## Number 1 · Working with fractions (Lesson 1.1)

**Worked example: multiplying fractions**

Work out $\frac{3}{4} \times \frac{8}{9}$

Give your answer in its simplest form.

| | |
|---|---|
| **Step 1** Multiply the tops | $3 \times 8 = 24$ |
| **Step 2** Multiply the bottoms | $4 \times 9 = 36$ |
| **Step 3** Simplify | $\frac{24}{36} = \frac{2}{3}$ |
| **Step 4** Answer | $\frac{2}{3}$ |

### Practice: multiplying fractions

**1.7** Work out $\frac{5}{6} \times \frac{3}{10}$. Give your answer in its simplest form.

**1.8** Work out $\frac{4}{7} \times \frac{7}{12}$. Give your answer in its simplest form.

**1.9** Work out $\frac{2}{5} \times \frac{15}{16}$. Give your answer in its simplest form.

**1.10** Work out $\frac{3}{4} \times \frac{2}{9}$. Give your answer in its simplest form.

**1.11** Work out $\frac{5}{8} \times \frac{4}{15}$. Give your answer in its simplest form.

**1.12** Work out $\frac{7}{9} \times \frac{3}{14}$. Give your answer in its simplest form.

## Number 1 · Working with fractions (Lesson 1.1)

**Worked example: dividing fractions**

Work out $\frac{5}{6} \div \frac{2}{3}$

Give your answer as a mixed number.

| | |
|---|---|
| **Step 1** Flip and multiply | $\frac{5}{6} \times \frac{3}{2} = \frac{15}{12}$ |
| **Step 2** Simplify | $\frac{5}{4}$ |
| **Step 3** Answer | $1\frac{1}{4}$ |

### Practice: dividing fractions

**1.13** Work out $\frac{3}{4} \div \frac{2}{5}$. Give your answer as a mixed number.

**1.14** Work out $\frac{7}{8} \div \frac{1}{3}$. Give your answer as a mixed number.

**1.15** Work out $\frac{9}{10} \div \frac{3}{5}$. Give your answer as a mixed number.

**1.16** Work out $\frac{5}{6} \div \frac{4}{9}$. Give your answer as a mixed number.

**1.17** Work out $\frac{2}{3} \div \frac{4}{7}$. Give your answer as a mixed number.

**1.18** Work out $\frac{8}{9} \div \frac{2}{3}$. Give your answer as a mixed number.

## Number 1 · Working with fractions (Lesson 1.1)

**Worked example: fractions of amounts**

Work out $\frac{2}{5}$ of 35

| | | |
|---|---|---|
| **Step 1** | Divide by the bottom | $35 \div 5 = 7$ |
| **Step 2** | Multiply by the top | $7 \times 2 = 14$ |
| **Step 3** | Answer | $14$ |

### Practice: fractions of amounts

**1.19** Work out $\frac{5}{6}$ of 42

**1.20** Work out $\frac{2}{9}$ of 81

**1.21** Work out $\frac{4}{5}$ of 75

**1.22** Work out $\frac{3}{7}$ of 56

**1.23** Work out $\frac{5}{8}$ of 96

**1.24** Work out $\frac{2}{3}$ of 57

## Number 1 · Working with fractions (Lesson 1.1)

**Worked example: fractions, decimals and percentages**

Write $\frac{3}{8}$

(a) as a decimal. (b) as a percentage.

| | | |
|---|---|---|
| **(a)** | **Step 1** Top ÷ bottom | $3 \div 8 = 0.375$ |
| **(b)** | **Step 1** × 100 | $0.375 \times 100 = 37.5\%$ |

### Practice: fractions, decimals and percentages

**1.25** Write $\frac{2}{5}$

(a) as a decimal. (b) as a percentage.

**1.26** Write $\frac{7}{20}$

(a) as a decimal. (b) as a percentage.

**1.27** Write $\frac{5}{8}$

(a) as a decimal. (b) as a percentage.

**1.28** Write $\frac{3}{4}$

(a) as a decimal. (b) as a percentage.

**1.29** Write $\frac{9}{25}$

(a) as a decimal. (b) as a percentage.

**1.30** Write $\frac{1}{8}$

(a) as a decimal. (b) as a percentage.

## Number 1 · Order of operations and rounding (Lessons 1.2, 1.3)

**Worked example: order of operations**

Work out

(a) $9 + 2 \times 6$ (b) $(10 - 4)^2 \div 3$

| | | | |
|---|---|---|---|
| **(a)** | **Step 1** | Multiply first | $2 \times 6 = 12$ |
| | **Step 2** | Then add | $9 + 12 = 21$ |
| **(b)** | **Step 1** | Brackets first | $10 - 4 = 6$ |
| | **Step 2** | Then the power | $6^2 = 36$ |
| | **Step 3** | Then divide | $36 \div 3 = 12$ |

### Practice: order of operations

**2.1** Work out

(a) $7 + 4 \times 5$ (b) $(9 - 3)^2 \div 4$

**2.2** Work out

(a) $20 - 3 \times 6$ (b) $(2 + 5)^2 - 9$

**2.3** Work out

(a) $12 + 18 \div 3$ (b) $(14 - 6)^2 \div 8$

**2.4** Work out

(a) $6 + 5 \times 3$ (b) $(11 - 5)^2 \div 9$

**2.5** Work out

(a) $30 - 4 \times 5$ (b) $(3 + 4)^2 - 10$

**2.6** Work out

(a) $8 + 24 \div 4$ (b) $(15 - 7)^2 \div 4$

## Number 1 · Order of operations and rounding (Lessons 1.2, 1.3)

**Worked example: rounding**

Round 6.3758

(a) correct to 1 decimal place. (b) correct to 3 significant figures.

| | | | |
|---|---|---|---|
| **(a)** | **Step 1** | Next digit is 7: round up | $6.4$ |
| **(b)** | **Step 1** | Next digit is 5: round up | $6.38$ |

### Practice: rounding

**2.7** Round 5.2739

(a) correct to 1 decimal place. (b) correct to 3 significant figures.

**2.8** Round 18.4615

(a) correct to 1 decimal place. (b) correct to 3 significant figures.

**2.9** Round 9.0847

(a) correct to 1 decimal place. (b) correct to 3 significant figures.

**2.10** Round 3.6582

(a) correct to 1 decimal place. (b) correct to 3 significant figures.

**2.11** Round 27.354

(a) correct to 1 decimal place. (b) correct to 3 significant figures.

**2.12** Round 4.9963

(a) correct to 1 decimal place. (b) correct to 3 significant figures.

## Algebra 1 · Substitution and simplifying (Lessons 2.1, 2.2, 2.3)

**Worked example: substitution**

$a = 5$ and $b = -3$

(a) Work out the value of $2a + b$ (b) Work out the value of $a^2 - 4b$

| | | | |
|---|---|---|---|
| **(a)** | **Step 1** | Put the numbers in | $2 \times 5 + (-3)$ |
| | **Step 2** | Answer | $7$ |
| **(b)** | **Step 1** | Put the numbers in | $5^2 - 4 \times (-3)$ |
| | **Step 2** | Answer | $25 + 12 = 37$ |

### Practice: substitution

**3.1** $a = 4$ and $b = -1$

(a) Work out the value of $3a + b$ (b) Work out the value of $a^2 - 2b$

**3.2** $a = 2$ and $b = -5$

(a) Work out the value of $5a + b$ (b) Work out the value of $a^2 - 1b$

**3.3** $a = 6$ and $b = -2$

(a) Work out the value of $2a + b$ (b) Work out the value of $a^2 - 3b$

**3.4** $a = 5$ and $b = -3$

(a) Work out the value of $2a + b$ (b) Work out the value of $a^2 - 4b$

**3.5** $a = 3$ and $b = -4$

(a) Work out the value of $5a + b$ (b) Work out the value of $a^2 - 2b$

**3.6** $a = 7$ and $b = -1$

(a) Work out the value of $2a + b$ (b) Work out the value of $a^2 - 5b$

## Algebra 1 · Substitution and simplifying (Lessons 2.1, 2.2, 2.3)

**Worked example: simplifying**

Simplify

(a) $6a + 3b - 2a + b$ (b) $5x \times 2y$

| | | | |
|---|---|---|---|
| **(a)** | **Step 1** | Collect $a$ | $6a - 2a = 4a$ |
| | **Step 2** | Collect $b$ | $3b + b = 4b$ |
| | **Step 3** | Answer | $4a + 4b$ |
| **(b)** | **Step 1** | Multiply the numbers | $5 \times 2 = 10$ |
| | **Step 2** | Answer | $10xy$ |

### Practice: simplifying

**3.7** Simplify

(a) $4x + 5y - x + 2y$ (b) $6a \times 3b$

**3.8** Simplify

(a) $7p - 2q - 3p + 6q$ (b) $2c \times 7d$

**3.9** Simplify

(a) $8a + b - 5a + 4b$ (b) $5m \times 4n$

**3.10** Simplify

(a) $9x + 2y - 4x + 3y$ (b) $4p \times 6q$

**3.11** Simplify

(a) $3m + 8n - m - 5n$ (b) $7x \times 2y$

**3.12** Simplify

(a) $6c + d - 2c + 4d$ (b) $3a \times 8b$

## Algebra 1 · Substitution and simplifying (Lessons 2.1, 2.2, 2.3)

**Worked example: expanding brackets**

Expand

(a) $3(4x - 2)$ (b) $x(x + 7)$

| | | | |
|---|---|---|---|
| **(a)** | **Step 1** | Multiply $4x$ by 3 | $3 \times 4x = 12x$ |
| | **Step 2** | Multiply $-2$ by 3 | $3 \times (-2) = -6$ |
| | **Step 3** | Answer | $12x - 6$ |
| **(b)** | **Step 1** | Multiply $x$ by $x$ | $x \times x = x^2$ |
| | **Step 2** | Multiply 7 by $x$ | $x \times 7 = 7x$ |
| | **Step 3** | Answer | $x^2 + 7x$ |

### Practice: expanding brackets

**3.13** Expand

(a) $4(3x - 5)$ (b) $x(x + 6)$

**3.14** Expand

(a) $6(2x + 1)$ (b) $x(x - 3)$

**3.15** Expand

(a) $2(5x - 4)$ (b) $x(x + 9)$

**3.16** Expand

(a) $7(2x - 3)$ (b) $x(x + 8)$

**3.17** Expand

(a) $5(3x + 1)$ (b) $x(x - 2)$

**3.18** Expand

(a) $3(4x - 7)$ (b) $x(x + 1)$

## Algebra 1 · Substitution and simplifying (Lessons 2.1, 2.2, 2.3)

**Worked example: multiplying the same letter**

Simplify $4p \times 3p$

| | | |
|---|---|---|
| **Step 1** | Multiply the numbers | $4 \times 3 = 12$ |
| **Step 2** | $p \times p$ | $p^2$ |
| **Step 3** | Answer | $12p^2$ |

### Practice: multiplying the same letter

**3.19** Simplify $3m \times 5m$

**3.20** Simplify $4x \times 2x$

**3.21** Simplify $6y \times y$

**3.22** Simplify $5a \times 4a$

**3.23** Simplify $2k \times 9k$

**3.24** Simplify $7n \times 3n$

## Algebra 1 · Solving equations (Lesson 2.4)

**Worked example: two-step equations**

Solve $4x + 7 = 31$

| | | |
|---|---|---|
| **Step 1** | Subtract 7 | $4x = 24$ |
| **Step 2** | Divide by 4 | $x = 6$ |

### Practice: two-step equations

**4.1** Solve $5x + 3 = 38$

**4.2** Solve $2x + 9 = 25$

**4.3** Solve $6x - 4 = 32$

**4.4** Solve $4x + 7 = 35$

**4.5** Solve $7x - 2 = 40$

**4.6** Solve $3x + 8 = 29$

## Algebra 1 · Solving equations (Lesson 2.4)

**Worked example: equations with brackets**

Solve $3(x-4)=18$

| | | |
|---|---|---|
| **Step 1** | Expand | $3x-12=18$ |
| **Step 2** | Add 12 | $3x=30$ |
| **Step 3** | Divide by 3 | $x=10$ |

### Practice: equations with brackets

**4.7** Solve $5(x-3)=20$

**4.8** Solve $2(x+6)=22$

**4.9** Solve $4(x-1)=28$

**4.10** Solve $3(x+5)=27$

**4.11** Solve $6(x-2)=30$

**4.12** Solve $2(x-7)=10$

**Worked example: unknowns on both sides**

Solve $6x - 5 = 2x + 15$

| | | |
|---|---|---|
| **Step 1** | Subtract $2x$ | $4x - 5 = 15$ |
| **Step 2** | Add 5 | $4x = 20$ |
| **Step 3** | Divide by 4 | $x = 5$ |

## Practice: unknowns on both sides

**4.13** Solve $5x - 2 = 2x + 13$

**4.14** Solve $8x + 1 = 3x + 26$

**4.15** Solve $9x - 7 = 4x + 8$

**4.16** Solve $6x + 4 = 2x + 24$

**4.17** Solve $7x - 5 = 3x + 11$

**4.18** Solve $5x + 3 = 2x + 18$

**Worked example: equations with x over a number**

Solve $\frac{x}{4} + 5 = 8$

| | | |
|---|---|---|
| **Step 1** | Subtract 5 | $\frac{x}{4} = 3$ |
| **Step 2** | Multiply by 4 | $x = 12$ |

## Practice: equations with x over a number

**4.19** Solve $\frac{x}{4} + 2 = 7$

**4.20** Solve $\frac{x}{5} + 3 = 9$

**4.21** Solve $\frac{x}{2} + 6 = 13$

**4.22** Solve $\frac{x}{6} + 1 = 4$

**4.23** Solve $\frac{x}{3} + 7 = 12$

**4.24** Solve $\frac{x}{5} - 2 = 4$